Check for updates

Received: 24 August 2023 | Accepted: 11 April 2024

DOI: 10.1002/nur.22388

RESEARCH ARTICLE

RESEARCH IN NURSING & HEALTH WILEY

# Validation of a short form of the practice environment scale of the nursing work index: The PES-5

Eileen T. Lake | Jennifer Gil | Lynne Moronski | Matthew D. McHugh | Linda H. Aiken | Karen B. Lasater

Department of Biobehavioral Health Sciences, Center for Health Outcomes and Policy Research, University of Pennsylvania School of Nursing, Philadelphia, Pennsylvania, USA

**Correspondence**
Eileen T. Lake, Center for Health Outcomes and Policy Research, University of Pennsylvania, Claire M. Fagin Hall, 418 Curie Boulevard Philadelphia, PA 19104-4217, USA.
Email: elake@nursing.upenn.edu

**Funding information**
Supported by funding from the National Institute of Nursing Research, Grant/Award Numbers: NINR - T32-NR-007104, NINR-2R01NR014855-06; and the National Council of State Boards of Nursing, Grant/Award Number: NCSBN CRE R201011

## Abstract

The Practice Environment Scale of the Nursing Work Index (PES-NWI) has been utilized for two decades globally to measure nurse work environments. Its 31 items in five domains present a substantial respondent burden, threatening survey response rates. The purpose of this study was to derive and validate a short form: the PES-5. We conducted a cross-sectional, secondary analysis of survey data from nurses in 760 hospitals in six U.S. states in 2016 or 2019. One representative item per subscale was selected by highest item-to-subscale $R^2$ from the original PES-NWI publication. Five psychometric properties of the PES-5 were evaluated. The reproduced structure of the full form was confirmed in the 2016 data by the highest $R^2$ for the selected items. The unidimensional structure of the PES-5 was confirmed through confirmatory factor analysis. The correlation between the composite values of the 28-item and 5-item versions was 0.94. The Cronbach's alpha reliability of the PES-5 was >0.80. The intraclass correlation coefficient (ICC 1, *k*), which evaluates the stability of aggregated values when data are clustered, i.e., nurses are nested within hospitals, was >0.80 in both datasets, demonstrating satisfactory aggregate properties. Construct validity was supported by the selected items being ranked highly in their respective subscales by an expert panel. Criterion validity was supported by an analysis of variance of the PES-5 mean value across responses to a single-item work environment measure. Similar patterns of relationships with other key variables were identified by statistically significant odds ratios in regression models predicting patient mortality from the PES-5. The classification accuracy of the PES-5 was high, with 88% of hospitals classified identically by both versions. The PES-5 shows promise for measurement of nurses' work environments while maximizing response rate by reducing participant burden.

KEYWORDS

nurses, psychometrics, surveys and questionnaires, work environment

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.
© 2024 The Authors. *Research in Nursing & Health* published by Wiley Periodicals LLC.

## 1 | INTRODUCTION

The Practice Environment Scale of the Nursing Work Index (PES-NWI) was developed in 2002 to measure the nursing practice environment so that managers and researchers could address the nursing shortage (Lake, 2002). The nursing practice environment was conceptualized from the disciplinary viewpoint of sociology of organizations, occupations, and work (Lake, 2002). This constellation of subdisciplines reveals the challenges inherent in organizing the work of a professional whose work entails treating patients in dynamic health states in a complex organization. These considerations underlie the phenomena of the nursing practice environment. Through a tool that could empirically measure the work environment, managers could address deficits and researchers could establish theorized relationships with quality, safety, and outcomes as well as providing benchmarks. The tool has been utilized for two decades globally to measure nurse work environments. The instrument comprises 31 items in five domains, which present substantial respondent burden, threatening response rate.

Several practical and theoretical gaps in this instrument have been identified in the literature. Lake (2007) evaluated the utility of various multidimensional instruments to measure the nursing practice environment. Utility was judged by theoretical relevance, ease of use, and dissemination. Although the PES-NWI was proposed as having the greatest utility, one recommendation was to develop a short form, specifically: "one or two items would be retained or developed to represent each key domain." (p. 118) Regarding theoretical gaps, Swiger et al.'s (2017) review noted that the contemporary relevance of the instrument content warranted investigation. The purpose of this study was to derive and validate a short form of the PES-NWI from the original instrument.

### 1.1 | Description of instrument

The PES-NWI was developed to measure the nursing practice environment, which was defined as the organizational traits that support or undermine professional nursing practice (Lake, 2002). It comprises 31 items in five subscales: (1) Nurse Participation in Hospital Affairs; (2) Nursing Foundations for Quality of Care; (3) Nurse Manager Ability, Leadership, and Support of Nurses; (4) Staffing and Resource Adequacy; and (5) Collegial Nurse–Physician Relations. For each item, the nurse responds to the query "Please indicate the extent to which you agree that the item is PRESENT IN YOUR CURRENT JOB." Four Likert-type responses are Strongly Disagree (value of 1), Disagree (2), Agree (3), and Strongly Agree (4). Subscale values are calculated as the mean of subscale item values. A composite is calculated as the mean of the subscale values. Higher values indicate better realization of a supportive work environment. The subscales were moderately correlated with each other (0.35–0.65, mean of 0.48) (Lake, 2002). The instrument has been continuously endorsed as a national quality standard since 2004 by the National Quality Forum (National Quality Forum, 2022) and now Battelle (Partnership for Quality Measurement, 2023).

Over two decades of use, the instrument has been used globally in research and management. It has been translated into 24 languages and 15 language variants in 35 countries (Lake et al., 2023). Several systematic reviews and a meta-analysis have summarized the empirical results (Havens et al., 2012; Lake et al., 2019; Swiger et al., 2017), which demonstrate consistent associations between nurses' work environments and quality of care, safety, patient satisfaction, nurse job outcomes, and patient health outcomes. The meta-analysis (Lake et al., 2019) included 17 studies representing 2677 hospitals, 165,024 nurses, and 1,368,420 patients, in 22 countries. A generalizability meta-analysis demonstrated high reliability across 51 studies including 68,278 nurses (Zangaro & Jones, 2019). The estimated reliability in 38 studies was 0.92. Nurse managers have used the instrument to measure and improve the work environment (Anderson et al., 2010; Nicholson et al., 2019).

### 1.2 | Shortened versions of the instrument

Few shortened versions of the instrument have been published. Montgomery et al. (2023) used item-response theory to reduce the instrument to achieve a parsimonious version. With nurse survey data from two different U.S. samples, they derived and then tested a 20-item version. The measurement model was confirmed in confirmatory factor analysis (CFA). High correlations (Kendall's Tau = 0.84–0.89) were reported between the original and reduced subscales in both samples. High reliability of the 20-item version was demonstrated in both samples.

Campbell et al. (2023) aimed to determine the relevance of the PES-NWI items to currently practicing nurses since being derived from the 1960s to 1980s and to determine whether additional items are needed to capture the contemporary context of nursing care. They employed a mixed-methods approach and utilized data from 818 nurses in 24 hospitals to test a modified 50-item PES-NWI version, comprising the 31 items plus 19 items related to technology, medical assistants, and items suggested by experts, plus three open-ended questions. Rather than asking the standard "this is present in my current job, they asked the two statements from the original Nursing Work Index (Kramer et al., 1989) for nurses to rate their agreement on how each item relates to their: (1) job satisfaction; and (2) ability to provide quality patient care. The findings of the study revealed that while the majority of items in the PES-NWI remained applicable to modern nursing practice, three could be removed and refinements in existing items or new content could enhance the instrument. We note that the implied survey length from their results would equal or exceed 47 items from the modified version tested and the recommended additional items.

In a study conducted by Djukic et al. (2021), investigators explored the factors linked to the implementation of evidence-based management practices by nurse managers to enhance nurse work environments. To achieve this, they utilized a modified version of the PES-NWI, which included 26 items of the original 31 items. The derivation of the 26 items was not reported. They reported the shortened version of the instrument's performance validated with confirmatory factors analysis in the study sample, comprising 331 nurse managers in 10 hospitals. Djukic,

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Jun, and Fletcher reported reliability coefficients of 0.80–0.95 for the reduced version and CFA validated the subscale factor structure of the instrument.

### 1.3 | Approaches to reducing instrument length

Various approaches have been proposed to reduce instrument length while retaining satisfactory measurement properties, such as Classical Test Theory and Item Response Theory (Kleka & Soroko, 2018). These approaches have been criticized, however, for yielding measures that do not reflect the underlying construct (Kleka & Soroko, 2018). Maloney et al. (2011) concluded that "a priori selection of items based on previous research may be a useful technique for developing short measures of psychological constructs that mitigate practical, analytical, and methodological issues associated with lengthy surveys." (p. 162). A goal in instrument reduction is to retain the original construct while reducing length and preserving reliability. They note that some items are strongly related to a construct and others moderately or weakly. They propose that researchers can retain the core construct by selecting the most representative items. Accordingly, they advise a systematic and conceptually sound selection method.

Kleka and Soroko (2018) note the two main issues are how to select items and how to assess the short form. They proposed a six-step standard abridgement procedure: (1) determine the desirable properties of a short form; (2) consider potential costs and benefits of reducing; (3) choose the statistical technique to select the items; (4) select the items; (5) describe the psychometric parameters; (6) evaluate the short form and confirm its psychometric properties. For step 6, they propose five ways to confirm the psychometric properties that the short form properly represents the measured construct (p. 157): reliability, validity (face and content), reproduced structure of the full form, similar patterns of relationships with other key variables, and similar classification accuracy of the short form against that of the full form instrument.

In the context of the post-pandemic nurse staffing crisis, the top recommendation is to reform the work environment (Nurse Staffing Task Force, 2023). Therefore, the significance of a low-burden survey measure of the work environment cannot be overstated. This widely used nurse survey instrument is a promising source for such a measure. Although several shorter forms of this instrument have been published, they all have at least 20 items. Utilizing approaches from survey research and psychology, the study purpose was to reduce the PES-NWI to achieve a valid and reliable short form.

## 2 | METHODS

### 2.1 | Design and data

This cross-sectional, secondary data analysis uses a single data source from 2016 and multiple linked data sources from 2019. Specifically, data from a 2016 large survey of registered nurses (RN4CAST) in California, Florida, New Jersey, and Pennsylvania was used to generate the 2016 sample from hospital-employed registered nurses' responses to the 28-item PES-NWI. Data from 2019 included a similar large survey of registered nurses in New York and Illinois to create the 2019 sample, as well as Medicare inpatient claims (MedPAR) data to evaluate the predictive validity of the short form instrument (i.e., PES-5) on in-hospital patient mortality. American Hospital Association Annual Survey data provided data on hospital characteristics used in risk-adjustment procedures and included hospital size, teaching status, and technology capabilities. The current study did not meet the regulatory definition of human research because it involved de-identified data that were previously collected as part of a research study by another investigator. The IRB provided a letter stating that the project did not require IRB review or approval.

### 2.2 | Samples

The 2016 sample was comprised of registered nurses who were employed in a hospital in one of four states: California, Florida, New Jersey, Pennsylvania. Hospitals were included in the sample if they had at least 10 nurse respondents reporting on the 28-item PES-NWI for that hospital, and if the hospital was an adult non-federal acute care hospital. This sample included 488 hospitals and 17,134 nurses.

The 2019 sample was comprised of registered nurses who were employed in a hospital in either New York or Illinois. Hospitals were included in the sample if they had at least 10 nurse respondents reporting on the PES-5 for that hospital, and if the hospital was an adult non-federal acute care hospital. This sample included 272 hospitals and 19,942 nurses.

The response rates to the surveys were 26% (2016 survey) and 18% (2019 survey). A nonrespondent survey established that there was no nonresponse bias in the work environment variable (Lasater et al., 2019). Of particular importance is hospital inclusion to evaluate nurse agreement within and between hospitals and to estimate the association of patient mortality to varying work environment across hospitals. While response rates were moderately low at the nurse-level, we have data from 98% of acute care general hospitals in both the 2016 and 2019 sampling frames that have more than 250 beds.

To evaluate the predictive validity of the PES-5, a sample of Medicare beneficiary inpatients was derived from MedPAR data. Patients were included if they had a hospital admission to one of the 272 New York or Illinois hospitals in 2019 and were between 65 and 99 years of age. Patients were excluded if their principal reason for admission was for a psychiatric or drug/alcohol abuse diagnosis. The final patient sample was 745,073 patients.

### 2.3 | Measures

The PES-NWI stem is the following: "For each item, please indicate the extent to which you agree that the item is PRESENT IN YOUR CURRENT JOB." An example item is "Adequate support services allow me to spend time with my patients." There are four Likert-type

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

response options: strongly disagree, disagree, agree, and strongly agree, which are scored as 1, 2, 3, and 4. Thus, higher scores indicate greater agreement that the desired organizational traits are present. The hypothetical range is 1–4 for each item and for the composite value, which is the mean of the five subscale values.

The two primary measures in this analysis were: the 28-item PES-NWI and the PES-5. The 28-item PES-NWI utilized herein was first reported in a 2008 publication (Aiken et al., 2008) and was utilized subsequently by these researchers, including the secondary data used for this study. Three omitted items were considered outdated (use of nursing diagnosis) or correlated with other key variables of interest (nursing model of care; preceptorship).

The PES-5 included one item from each of the 5 subscales. The five items were selected for the PES-5 instrument because they had the highest $R^2$ in the prior published work on the PES-NWI (Lake, 2002) using the 1999 sample. The 5 items are: (1) Administration that listens and responds to nurse concerns, (2) A supervisor who is a good manager and leader, (3) Good teamwork between nurses and physicians, (4) Enough staff to get work done, (5) A clear philosophy of nursing that pervades the patient care environment.

In the regression analyses, the patient outcome studied was in-hospital mortality, defined as a death occurring during the hospitalization. Covariates in the model included patient characteristics (i.e., age, sex, transfer status, an Elixhauser comorbidity index score, base DRG dummies) and hospital characteristics (i.e., bed size category, teaching status, technology capabilities).

## 2.4 | Analysis

To derive and validate the PES-5 instrument, we followed the six-step abridgement procedure (Kleka & Soroko, 2018). Regarding potential costs and benefits, we let the benefits of high response rate determine the shortest form that would still retain conceptual integrity, i.e., the original subscale domains. The potential cost of reducing an instrument is reduced construct validity. In this instance, we expected construct validity to remain intact due to selecting one item from each subscale. Therefore, we judged that the benefits outweighed the costs. Our approach was to select one item from each subscale that had the highest explained variance ($R^2$) of the subscale score from the original publication using the 1999 sample (Lake, 2002). We then confirmed, using the 2016 sample, that those items also had the highest explained variance ($R^2$) of the subscale score. We first report reproduced structure because dimensionality should precede reliability consideration. That is, Cronbach's alpha is only applicable to unidimensional scales and should not be computed before unidimensionality is established via CFA.

### 2.4.1 | Reproduced structure of the full form

The reproduced structure of the full form was evaluated by (1) the highest $R^2$ for the selected items in the 2016 data and (2) CFA of the five-item version. The purpose of the CFA was to confirm that the five-item version was a unidimensional representation of the original 31-item version PES-NWI. Specifically, the CFA confirmed that the five-item version represented a single factor reflecting a work environment measure. A satisfactory model fit is suggested root mean squared error of approximation (RMSEA) values less than 0.08 (Browne & Cudeck, 1992), comparative fit index (CFI) values bigger than 0.95, and NFI greater than 0.90 (Byrne, 2013). We calculated the Tucker-Lewis Index (TLI), a variation of the NFI. The fit indices (CFI, TLI, and RMSEA) are considered robust to the potential large sample inflation effect, which occurs when there are few observations, in this case the five PES items, in the context of a large sample. The CFA was conducted in STATA using the sem (structural equation modeling) command with the adf (asymptotic distribution free) method.

### 2.4.2 | Evaluation of reliability

Two types of reliability of the PES-5 were calculated in both datasets: Cronbach's alpha, a measure of internal consistency, and the intraclass correlation coefficient, ICC (1, *k*), which reflects the stability of the PES-5 score for clustered data, i.e., nurses within a hospital. The ICC (1, *k*) is used to establish that an instrument demonstrates satisfactory aggregate properties and was computed using a one-way analysis of variance (ANOVA) test in which we reported the estimated reliability of the hospital mean. Specifically, we computed the Cronbach's alpha and the ICC (1, *k*) in the 2019 sample and compared it with those of the same 5-items and the 28-items in the 2016 sample.

### 2.4.3 | Evaluation of validity

We evaluated the content validity of the 5-item version by recruiting an expert panel to rank items within subscales by their importance to the subscale domain to determine if the selected items rank highly. The panel comprised three nurses with at least 2 years of clinical experience who had undergraduate research assistant experience with the PES-NWI and five nurse researchers with extensive experience with the PES-NWI. All were asked to rank items by their relevance to the domain. To explore the construct validity of the score of the short form to the original, we correlated the two values in the 2016 data.

We evaluated criterion-related validity by conducting an ANOVA from both samples on the short form score with responses to a single item question: "How would you rate the organizational work environment of your job (e.g., adequacy of resources, relationships with coworkers, support from supervisors)?" The response categories were excellent, good, fair, and poor.

### 2.4.4 | Similar patterns of relationships with other key variables

Similar patterns of relationships with other key variables were explored in multi-level logistic regression models which accounted

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

for clustering of patients within hospitals to predict in-patient mortality (measured at the patient-level) from the PES-5 (measured at the hospital-level) in 2019. In these models, to ease interpretation of the results, the PES items were standardized by subtracting the overall mean from the item mean to center at zero and then divided by the standard deviation, such that a one unit change in the score could be interpreted as a 1 standard deviation change.

#### 2.4.5 | Similar classification accuracy of the short form against that of the full form instrument

The classification accuracy of the PES-5 was tested in the 2016 data by classifying hospitals into better, mixed, and poor work environments using the 28-item instrument and the PES-5. This classification was based on a hospital's composite score calculated from the subscale means in the 28-item instrument and the means in the PES-5. These composite scores were arrayed and hospitals in the top quartile, the middle 50%, and the bottom quartile were identified as having better, mixed, and poor work environments. Krippendorff's alpha was calculated as a measure of classification accuracy. The percentage of hospitals matched identically by the 28-item and the 5-item instruments was calculated.

## 3 | RESULTS

In Table 1 we report the $R^2$ of the 28-items with their corresponding subscales in the 2016 sample as well as the $R^2$ of the same 28-items from data collected in 1999 and previously published in Lake (2002). Using the 1999 sample, the items with the highest $R^2$ within each subscale were selected as the 5-items for the PES-5. In Table 1, we confirm that in all the five subscales, except Staffing and Resource Adequacy, the items with the highest $R^2$ in the 1999 data are also the items with the highest $R^2$ in the 2016 sample. In the Staffing and Resource Adequacy subscale, the item with the highest $R^2$ in the 1999 data, which is the item selected for inclusion in the PES-5, was "Enough staff to get the work done" ($R^2$ 0.71). The $R^2$ for this item in the 2016 sample was 0.81, only slightly lower than the highest performing item "Enough registered nurses to provide quality patient care" ($R^2$ 0.82).

### 3.1 | Reliability

To evaluate the reliability of the 28-item PES-NWI and the PES-5, Cronbach's alphas are reported (Table 2). The 28-item PES-NWI in the 2016 sample had the highest Cronbach's alpha (0.96) indicating excellent internal consistency. The PES-5 in both samples performed equally well—0.82 and 0.81, respectively—indicating good internal consistency. ICC (1, *k*) reflected good stability of the PES-5 when nurse reports were aggregated to the hospital level. In the 2016 and 2019 samples, ICC (1, *k*) scores were 0.81 and 0.89, respectively.

In Table 3, the hospital-level means and standard deviations of the 28-item PES-NWI, each of the five subscales, and each item of the PES-5 are reported in the 2016 sample. The PES-5 hospital-level means and *SD*s are reported for the 2019 sample. The 28-item PES-NWI score in the 2016 sample was 2.80, reflecting a tendency toward "agree" (value of 3.0) that the traits are present in the current job. As is generally the pattern, the lowest scores were for Staffing and Resource Adequacy (2.51) and the highest for Nursing Foundations for Quality of Care (3.05). Regarding the PES-5, the lowest item in both samples was "Enough staff to get the work done" consistent with its Staffing and Resource Adequacy subscale. The highest values were for "A clear philosophy of nursing…" and "A lot of teamwork…," reflecting the source subscales whose values were the two highest in the 2016 sample.

The number and characteristics of the hospitals in both samples are described in Table 4. The 2016 sample was comprised of 488 hospitals in 2016. A total of 40% of these hospitals were located in California, the others in Florida (26%), Pennsylvania (22%) and New Jersey (11%). Most hospitals in the 2016 sample had more than 250 beds, were minor teaching hospitals, and had high technology capabilities. Comparatively, the 2019 sample was comprised of 272 hospitals in 2019, 53% of which were located in New York and 47% in Illinois. These hospitals were generally smaller than the 2016 sample hospitals, most did not have high technology capabilities, and the greatest percentage of them were not teaching hospitals.

### 3.2 | Validity

The results of the expert panel ranking were that all five selected items were ranked as first or second within their respective subscale, confirming single-item construct validity. The ANOVA resulted in significantly different values in a stair-step pattern for the five-item measure score across the four categories (excellent to poor). Both *p* values were <0.001. In both samples, nurses who rated their work environment as excellent had values on the five-item measure reflecting between agree and strongly agree that the items were present. The converse was also evident. Nurses rating their work environment as poor had values reflecting between disagree and strongly disagree. The correlation between the composite values of the 28-item and 5-item versions was 0.94.

The CFA results confirmed the unidimensional structure of the five items as follows: the RMSEA = 0.031, the CFI = 0.987, and the TLI = 0.975. The model versus saturated Chi-square was 908.04 ($p < 0.001$) and the baseline versus saturated was 19548.98 ($p < 0.001$). All indices met proper thresholds indicating satisfactory model fit.

The PES-5 in the 2019 sample demonstrated strong predictive validity (Table 5). Specifically, risk-adjusted regression models of the 5 items individually, as well as the average score of the PES-5, demonstrated statistically significant associations with in-hospital mortality, with odds ratios ranging from 0.85 to 0.92. That is, an increase of one *SD* was associated with 8%–15% lower odds that a patient would die in hospital. Given the *SD*s range from 0.25 to 0.36, a one-unit increase is equivalent to three to four *SD*s, which is larger than typical variation.

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**TABLE 1** Explained variance ($R^2$) of PES-NWI items with their corresponding subscales as published in prior work from the 1999 sample and 2016 sample.

| | $R^2$ in 1999 sample Lake (2002) | $R^2$ in 2016 sample |
|---|---|---|
| **Nurse participation in hospital affairs** | | |
| Staff nurses are involved in the internal governance of the hospital. | 0.41 | 0.67 |
| Opportunity for staff nurses to participate in policy decisions. | 0.47 | 0.67 |
| Opportunities for advancement. | 0.47 | 0.64 |
| Administration that listens and responds to employee concerns. | **0.49** | **0.67** |
| A chief nursing officer who is highly visible and accessible to staff. | 0.39 | 0.56 |
| Career development/clinical ladder opportunity. | 0.37 | 0.57 |
| Staff nurses have the opportunity to serve on hospital and nursing committees | 0.33 | 0.62 |
| A chief nurse officer equal in power and authority to other top level hospital executives. | 0.26 | 0.42 |
| **Nursing foundations for quality of care** | | |
| An active quality assurance program. | 0.39 | 0.61 |
| A preceptor program for newly hired RNs. | 0.37 | 0.51 |
| Nursing care is based on a nursing, rather than a medical model. | 0.43 | 0.54 |
| Patient care assignments that foster continuity of care, i.e., the same nurse cares for the patient from 1 day to the next. | 0.21 | 0.52 |
| A clear philosophy of nursing that pervades the patient care environment. | **0.53** | **0.63** |
| Written, up-to-date nursing care plans for all patients. | 0.28 | 0.49 |
| High standards of nursing care are expected by the administration. | 0.35 | 0.35 |
| Active staff development or continuing education programs for nurses. | 0.32 | 0.53 |
| Working with nurses who are clinically competent. | 0.29 | 0.36 |
| **Nurse manager ability, leadership, and support** | | |
| A nurse manager who is a good manager and leader. | **0.69** | **0.78** |
| A nurse manager who backs up the nursing staff in decision making, even if the conflict is with a physician. | 0.64 | 0.78 |
| A supervisory staff that is supportive of the nurses. | 0.63 | 0.73 |
| Praise and recognition for a job well done. | 0.53 | 0.66 |
| **Staffing and resource adequacy** | | |
| Enough staff to get the work done. | **0.71** | 0.81 |
| Enough registered nurses to provide quality patient care. | 0.69 | **0.82** |
| Adequate support services allow me to spend time with my patients. | 0.64 | 0.71 |
| Enough time and opportunity to discuss patient care problems with other nurses. | 0.52 | 0.64 |
| **Collegial nurse-physician relationships** | | |
| A lot of teamwork between nurses and physicians. | **0.76** | **0.82** |
| Physicians and nurses have good working relationships. | 0.62 | 0.73 |
| Collaboration (joint practice) between nurses and physicians | 0.66 | 0.81 |

*Note*: Analyses were conducted at the nurse level. Bold values indicate the item with the highest explained variance ($R^2$) within the subscales for each of the surveys.

Abbreviation: PES-NWI, Practice Environment Scale of the Nursing Work Index.

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**TABLE 2** Cronbach's alphas and intra-class correlation coefficient in the 2016 and 2019 samples.

| | 2016 Sample | | 2019 Sample |
|---|---|---|---|
| Number of hospitals | 488 | | 272 |
| Number of nurses | 17,134 | | 19,942 |
| | 28-item | 5-item | 5-item |
| Cronbach's alpha | 0.96 | 0.82 | 0.81 |
| ICC (1, *k*) | 0.83 | 0.81 | 0.89 |

**TABLE 3** PES-NWI statistics in the 2016 and 2019 samples.

| | 2016 Sample | 2019 Sample |
|---|---|---|
| Number of hospitals | 488 | 272 |
| Number of nurses | 17,134 | 19,942 |
| Average number of nurses per hospital, mean (*SD*) | 35.1 (28.8) | 73.3 (87.3) |
| | Mean (SD) | Mean (SD) |
| 28-item PES-NWI | 2.80 (0.24) | – |
| 28-item PES-NWI by subscale | | |
| Nurse participation in hospital affairs | 2.66 (0.33) | – |
| Nursing foundations for quality of care | 3.05 (0.24) | – |
| Nurse manager ability, leadership, and support | 2.74 (0.28) | – |
| Staffing resource adequacy | 2.51 (0.31) | – |
| Collegial nurse-physician relationships | 3.03 (0.22) | – |
| 5 items PES-5 | 2.69 (0.27) | 2.75 (0.28) |
| Administration that listens and responds to nurse concerns | 2.38 (0.35) | 2.54 (0.34) |
| A clear philosophy of nursing that pervades the patient care environment | 2.98 (0.32) | 2.85 (0.33) |
| A nurse manager who is a good manager and leader | 2.80 (0.30) | 2.91 (0.30) |
| Enough staff to get work done | 2.34 (0.36) | 2.34 (0.40) |
| A lot of teamwork between nurses and physicians | 2.94 (0.25) | 3.12 (0.23) |

*Note*: The data presented here are based on hospital-level reports.

Abbreviation: PES-NWI, Practice Environment Scale of the Nursing Work Index.

Table 6 presents the crosstabulation of hospitals in the 2016 sample, classified into better, mixed, and poor nurse work environments using both the 28-item PES-NWI and the PES-5. Here we see few observations (12%) in off-diagonal cells. Visualized in Figure 1, it is evident that there is no bias in the misclassification of the 12%.

**TABLE 4** Hospital characteristics in the 2016 and 2019 samples.

| | 2016 Sample | 2019 Sample |
|---|---|---|
| Number of hospitals | 488 | 272 |
| Size | | |
| Small (≤100 beds) | 24 (4.9%) | 63 (23.2%) |
| Medium (101–250 beds) | 199 (40.8%) | 92 (33.8%) |
| Large (>250 beds) | 365 (54.3%) | 117 (43.0%) |
| Teaching status | | |
| None | 203 (41.6%) | 116 (45.7%) |
| Minor | 238 (48.8%) | 85 (33.5%) |
| Major | 47 (9.6%) | 53 (20.9%) |
| Technology capabilities | | |
| Non-high | 212 (43.4%) | 128 (59.5%) |
| High | 276 (56.6%) | 87 (49.5%) |
| State | | |
| California | 198 (40.6%) | – |
| Florida | 127 (26.0%) | – |
| New Jersey | 55 (11.3%) | – |
| Pennsylvania | 108 (22.1%) | – |
| New York | – | 144 (52.9%) |
| Illinois | – | 128 (47.1%) |

*Note*: Technology capability was defined by whether or not the hospital had capabilities to perform open-heart surgery and/or major organ transplantation.

**TABLE 5** PES-5 item coefficients predicting risk-adjusted in-hospital patient mortality in the 2019 sample.

| | Odds ratio (95% CI) | *p* value |
|---|---|---|
| Administration that listens and responds to nurse concerns | 0.85 [0.78–0.92] | <0.001 |
| A supervisor who is a good manager and leader | 0.86 [0.80–0.93] | <0.001 |
| Good teamwork between nurses and physicians | 0.92 [0.85–0.99] | 0.035 |
| Enough staff to get work done | 0.86 [0.80–0.94] | 0.001 |
| A clear philosophy of nursing that pervades the patient care environment | 0.85 [0.78–0.92] | <0.001 |
| **PES-5** | 0.85 [0.78–0.93] | <0.001 |

*Note*: Risk-adjusted odds ratio reflect controls for patient characteristics (i.e., age, sex, transfer status, an Elixhauser comorbidity index score, base DRG dummies) and hospital characteristics (i.e., bed size category, teaching status, technology capabilities).

Abbreviation: CI, confidence interval.

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**TABLE 6** Classification accuracy of hospitals into Poor, Mixed, Better nurse work environments in the 2016 sample.

| | PES-5 | | | |
|---|---|---|---|---|
| | Poor | Mixed | Better | Total |
| 28-item PES-NWI | | | | |
| **Poor** | 108 | 14 | 0 | **122** |
| **Mixed** | 14 | 215 | 15 | **244** |
| **Better** | 0 | 15 | 107 | **122** |
| **Total** | **122** | **244** | **122** | **488** |

*Note*: Percent agreement: 88.11% ($p < 0.001$). Krippendorff's alpha coefficient 0.8100 ($p < 0.001$).

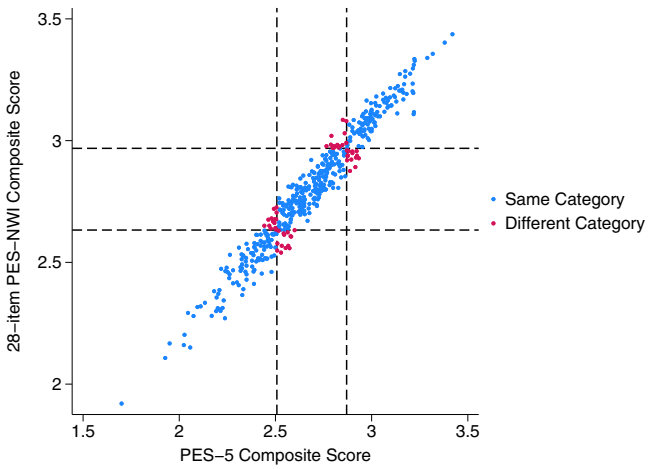


**FIGURE 1** Scatterplot of the 28-item and 5-item PES-NWI composite scores in the 2016 hospital sample. PES-NWI, Practice Environment Scale of the Nursing Work Index.

That is, these 12% are nearly equally distributed across the adjacent cell to the full form classification. Krippendorff's alpha was 0.81. The percent agreement was 88%.

## 4 | DISCUSSION

We were motivated to derive a short form of the PES-NWI to ease survey burden while maintaining construct validity. We utilized the original published statistics from the 1999 sample to select representative items for each domain and evaluated their psychometric properties in two recent large samples of nurses from 2016 to 2019. The selected items for the short form instrument (i.e., PES-5) demonstrated high reliability (both internal consistency and as an organization-level aggregate), construct and criterion validity, reproduced structure (by high $R^2$ with subscale and by CFA), expected patterns of relationships with key variables, and high classification accuracy.

The Cronbach's alphas of 0.81 on average is considered high reliability (DeVon et al., 2007). These values are lower than the mean score reliability of 0.92 reported by Zangaro for the full instrument (Zangaro & Jones, 2019), which is not surprising given the PES-5 has only five items. The high reliability, calculated in two large contemporary samples, is promising for the purpose of ongoing broad use of the instrument with considerably less respondent burden.

The high intraclass correlation coefficients, on average 0.85, indicate that work environment ratings of nurses within the same hospital are more similar than those of nurses across hospitals. The minimum threshold for aggregation of responses to the group level is 0.60 (Glick, 1985). Therefore, the PES-5 exhibits a clear capacity to differentiate the work environments of various hospitals, which is a key goal in much research with the full instrument. This ICC (1, *k*) is lower than the values reported in the original publication (Lake, 2002) with the full instrument, 0.96. The lower value is expected due to an instrument being reduced by 84% (31 − 5 = 26 items omitted; 26/31 items = 84%).

The structure of the original instrument was reproduced by four of the five items having the highest $R^2$ in the 2016 sample, and the other one having an $R^2$ within 0.01 of the highest value in the 2016 sample. Although it might be considered optimal to have the top five items with explained variance in their respective subscales, several facts support the suitability of items chosen from the original publication. First, each sample will yield differing results. Second, in the case of the Staffing and Resource Adequacy Subscale, the selected item's $R^2$ value was trivially different from the highest item in its subscale. Third, the items' performance was robust. It is likely that in future samples, the results would again vary. Additionally, the CFA of the five items confirmed satisfactory model fit, supporting unidimensional structure.

In our assessment of expected patterns of relationships with key variables, we found that each PES-5 item and the five items collectively predicted patient odds of in-hospital mortality at statistically significant levels. The odds ratios were, 0.85–0.92, and the overall set had an odds ratio of 0.85. That is, a one *SD* increase in each item or the composite was associated with 8%–15% lower odds of mortality for hospitalized elderly patients. These odds ratios appear similar to those reported previously, but several factors make direct comparison difficult. For example, a meta-analysis of three studies (Lake et al., 2019) estimated 8% lower odds of 30-day mortality among patients hospitalized for common surgeries for patients in better work environments, as compared to poor or mixed *or* poor work environments classified based on PES-NWI subscale scores exceeding the sample median or one *SD* higher PES-NWI composite score (Aiken et al., 2008, 2011; Cho et al., 2015). A similar study of 40 public hospitals in Chile found a similar odds ratio of 0.95, i.e., 5% lower odds, which was not significant (Aiken et al., 2021).

The PES-5 achieved excellent classification accuracy. Eighty-eight percent of hospitals were classified identically by the PES-5 and the remainder were classified in an adjacent category. As accurate classification is an evaluation of reliability, we refer to a threshold of 0.80 (or 80%) as reflecting high accuracy (DeVon et al., 2007). We note for researchers and managers that the thresholds in Figure 1 may be useful to identify better, mixed, and poor environments. These reflect this sample's distribution, which is broadly representative of U.S. hospitals.

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

### 4.1 | Implications of findings

Our findings have implications for research, practice, and policy. For consistent evidence from future research, we advocate utilizing these same five items for comparison across studies. Use of the original instrument is warranted when distinct subscale values are desired. For practice, the PES-5 simplifies the routine assessment of the work environment by managers and likely increases staff nurse participation, but again, would not permit particular subscales to be measured. From a policy perspective, the PES-5 could increase the willingness of federal, state, accreditation, or other regulatory authorities to require reporting by hospitals or other healthcare settings. For example, the state of Colorado requires hospitals with at least 200 beds to survey nurses annually with the complete PES-NWI (Colorado Hospital Quality Report by Measure, 2022). Also, the Leapfrog Group currently uses four measures to assess nursing workforce (The Leapfrog Group, 2022) and could add nurse surveys using the PES-5 to its requirements. In the context of pervasive surveying in our society, the PES-5 can alleviate survey burden while still providing crucial data for managers and policymakers. Thus, the practical significance and applicability of the PES-5 are wide-ranging, from single-institution dashboards to regulatory body oversight.

### 4.2 | Limitations

Selecting the five representative items on the basis of data published two decades ago may have resulted in a short form with less relevance to current practice. Our results from the 2016 data, however, demonstrated the ongoing dominance of the selected items to their respective subscales. Our expert panelists, however, endorsed the current relevance of the items they nominated to represent the subscales. Our approach assumed the domains measured in the subscales themselves continue to demonstrate their relevance and alignment with the current nurse work environments. This assumption is supported by results from a study of 818 nurses from 24 hospitals in 2021 on nurses' perspectives on the PES-NWI, in which the authors concluded that most items remain relevant for modern nursing practice (Campbell et al., 2023). The 2016 and 2019 nurse samples may not have represented nurses throughout the United States. Nevertheless, the six states represented are highly populous and geographically diverse.

## 5 | CONCLUSION

The PES-5 demonstrates satisfactory psychometric properties. The PES-5 shows promise for measurement of nurses' work environments while maximizing response rate by reducing participant burden. Additional research in non-hospital settings will broaden the evidence regarding its psychometric performance.

### AUTHOR CONTRIBUTIONS

All authors of this study substantially contributed to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work, drafted the work or reviewed it critically for important intellectual content, provided final approval of the version to be published, and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

### ACKNOWLEDGMENTS

Research for this article was conducted at the Center for Health Outcomes and Policy Research, University of Pennsylvania School of Nursing, Philadelphia, PA. The authors are thankful to analyst, Tim Cheney. Supported by funding from the National Institute of Nursing Research (NINR - T32-NR-007104; NINR-2R01NR014855-06) and the National Council of State Boards of Nursing (NCSBN CRE R201011).

### CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

### DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

### ORCID

*Eileen T. Lake* http://orcid.org/0000-0002-8823-3436
*Jennifer Gil* http://orcid.org/0009-0009-9309-2082
*Lynne Moronski* http://orcid.org/0009-0002-5875-4203
*Karen B. Lasater* http://orcid.org/0000-0002-5834-1954

### PEER REVIEW

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1002/nur.22388.

### REFERENCES

Aiken, L. H., Cimiotti, J. P., Sloane, D. M., Smith, H. L., Flynn, L., & Neff, D. F. (2011). Effects of nurse staffing and nurse education on patient deaths in hospitals with different nurse work environments. *Medical Care*, *49*(12), 1047–1053. https://doi.org/10.1097/MLR.0b013e3182330b6e

Aiken, L. H., Clarke, S. P., Sloane, D. M., Lake, E. T., & Cheney, T. (2008). Effects of hospital care environment on patient mortality and nurse outcomes. *JONA: The Journal of Nursing Administration*, *38*(5), 223–229. https://doi.org/10.1097/01.NNA.0000312773.42352.d7

Aiken, L. H., Simonetti, M., Sloane, D. M., Cerón, C., Soto, P., Bravo, D., Galiano, A., Behrman, J. R., Smith, H. L., McHugh, M. D., & Lake, E. T. (2021). Hospital nurse staffing and patient outcomes in Chile: A multilevel cross-sectional study. *The Lancet Global Health*, *9*(8), e1145–e1153.

Anderson, B. J., Manno, M., O'Connor, P., & Gallagher, E. (2010). Listening to nursing leaders: Using national database of nursing quality indicators data to study excellence in nursing leadership. *JONA: The Journal of Nursing Administration*, *40*(4), 182–187.

Browne, M., & Cudeck, R. (1992). Alternative ways of assessing model fit. *Sociological Methods & Research*, *21*(2), 230–258.

Byrne, B. M. (2013). *Structural equation modeling with EQS: Basic concepts, applications, and programming*. Routledge.

Campbell, C. M., Swiger, P. A., Warshawsky, N., Li, P., Olds, D., Cramer, E., & Patrician, P. A. (2023). Measuring the work environment: The

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

voice of the staff nurse. *JONA: The Journal of Nursing Administration*, *53*(5), 284–291.

Cho, E., Sloane, D. M., Kim, E.-Y., Kim, S., Choi, M., Yoo, I. Y., Lee, H. S., & Aiken, L. H. (2015). Effects of nurse staffing, work environments, and education on patient mortality: An observational study. *International Journal of Nursing Studies*, *52*(2), 535–542.

Colorado Hospital Quality Report by Measure. (2022). Quality and Nurse Sensitive Measures Report by Hospital. https://www.cohospitalquality.org/corda/dashboards/COLORADO_REPORT_CARD_BY_HOSPITAL/main.dashxml

DeVon, H. A., Block, M. E., Moyle-Wright, P., Ernst, D. M., Hayden, S. J., Lazzara, D. J., Savoy, S. M., & Kostas-Polston, E. (2007). A psychometric toolbox for testing validity and reliability. *Journal of Nursing Scholarship*, *39*(2), 155–164.

Djukic, M., Jun, J., & Fletcher, J. (2021). An examination of the factors associated with implementation of evidence-based management practices for improving nurse work environments. *Worldviews on Evidence-Based Nursing*, *18*(2), 129–137. https://doi.org/10.1111/wvn.12497

Glick, W. H. (1985). Conceptualizing and measuring organizational and psychological climate: Pitfalls in multilevel research. *The Academy of Management Review*, *10*(3), 601–616.

Havens, D. S., Warshawsky, N., & Vasey, J. (2012). The nursing practice environment in rural hospitals: Practice environment scale of the nursing work index assessment. *JONA: The Journal of Nursing Administration*, *42*(11), 519–525.

Kleka, P., & Soroko, E. (2018). How to avoid the sins of questionnaires abridgement. Survey Research Methods.

Kramer, M., Schmalenberg, C., & Hafner, L. P. (1989). What causes job satisfaction and productivity of quality nursing care? In T. F. Moore & E. A. Simendinger (Eds.), *Managing the nursing shortage: A guide to recruitment and retention* (pp. 12–32).

Lake, E. T. (2002). Development of the practice environment scale of the nursing work index. *Research in Nursing & Health*, *25*(3), 176–188. https://doi.org/10.1002/nur.10032

Lake, E. T. (2007). The nursing practice environment. *Medical Care Research and Review*, *64*(2), 104S–122S.

Lake, E. T., Rosenbaum, K. E. F., Sauveur, C., Buren, C., & Cho, P. (2023). Translations of the practice environment scale of the nursing work index: A systematic review. *Nursing & Health Sciences*, *25*(3), 365–380. https://doi.org/10.1111/nhs.13034

Lake, E. T., Sanders, J., Duan, R., Riman, K. A., Schoenauer, K. M., & Chen, Y. (2019). A meta-analysis of the associations between the nurse work environment in hospitals and 4 sets of outcomes. *Medical Care*, *57*(5), 353–361. https://doi.org/10.1097/MLR.0000000000001109

Lasater, K. B., Jarrín, O. F., Aiken, L. H., McHugh, M. D., Sloane, D. M., & Smith, H. L. (2019). A methodology for studying organizational performance: A multistate survey of front-line providers. *Medical Care*, *57*(9), 742–749. https://doi.org/10.1097/MLR.0000000000001167

Maloney, P., Grawitch, M. J., & Barber, L. K. (2011). Strategic item selection to reduce survey length: Reduction in validity? *Consulting Psychology Journal: Practice and Research*, *63*(3), 162–175.

Montgomery, A. P., Campbell, C. M., Azuero, A., Swiger, P. A., & Patrician, P. A. (2023). Using item response theory to develop a shortened practice environment scale of the nursing work index. *Research in Nursing & Health*, *46*(4), 400–410. https://doi.org/10.1002/nur.22324

National Quality Forum. (2022). *Practice Environment Scale - Nursing Work Index (PES-NWI) (composite and five subscales)*. National Quality Forum. Retrieved June 21 from http://www.qualityforum.org/QPS/3450

Nicholson, L., Ortiz, M. V., Wang, Y., Walsh, H., Ottolini, M. C., & Agrawal, D. (2019). Successful implementation of a novel collaborative interprofessional educational curriculum for nurses and residents in a pediatric acute care setting. *Journal of Interprofessional Education & Practice*, *17*, 100284.

Nurse Staffing Task Force. (2023). *Nurse Staffing Task Force Imperatives, Recommendations, and Actions*.

Partnership for Quality Measurement. (2023). *Practice Environment Scale - Nursing Work Index (PES-NWI) (composite and five subscales)*. Battelle. Retrieved July 11 from https://p4qm.org/measures?combine=3450

Swiger, P. A., Patrician, P. A., Miltner, R. S., Raju, D., Breckenridge-Sproat, S., & Loan, L. A. (2017). The practice environment scale of the nursing work index: An updated review and recommendations for use. *International Journal of Nursing Studies*, *74*, 76–84.

The Leapfrog Group. (2022). Support for the Nursing Workforce. https://ratings.leapfroggroup.org/measure/hospital/2022/support-nursing-workforce

Zangaro, G. A., & Jones, K. (2019). Practice environment scale of the nursing work index: A reliability generalization meta-analysis. *Western Journal of Nursing Research*, *41*(11), 1658–1684.

**How to cite this article:** Lake, E. T., Gil, J., Moronski, L., McHugh, M. D., Aiken, L. H., & Lasater, K. B. (2024). Validation of a short form of the practice environment scale of the nursing work index: The PES-5. *Research in Nursing & Health*, *47*, 450–459. https://doi.org/10.1002/nur.22388

1098240x, 2024, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nur.22388 by University Of Pennsylvania, Wiley Online Library on [21/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License